Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/30/2011
PI Name:	Held, Kathryn Ph.D.		
Project Title:	Mechanisms for Induction of Bystander Effects by High Energy Particles in Cells and Tissues		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kheld@partners.org	**Fax:**	FY 617-724-8320
PI Organization Type:	NON-PROFIT	**Phone:**	617-726-8161
Organization Name:	Massachusettes General Hospital		
PI Address 1:	Radiation Oncology		
PI Address 2:	55 Fruit Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114-2696	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	07/01/2007	**End Date:**	12/31/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: NCE to 12/31/2011 awarded (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX07AE40G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In space, astronauts are exposed to a complex and unique radiation environment composed of trapped proton and electron radiation, galactic cosmic rays (GCR; energetic protons and heavy charged ions, HZE), and solar particle events (SPE), primarily protons. Although these radiations can result in appreciable cumulative doses to astronauts during long-duration missions, the exposures occur at low fluences and fluence rates, such that particle traversals through individual cells in an astronaut's body may be well separated in tissue location and time. Under such conditions, so-called non-targeted effects such as bystander responses may have increased significance. The radiation-induced bystander effect is the occurrence of biological responses in unirradiated cells near to or sharing medium with cells actually traversed by radiation. We hypothesize that at the low fluences of protons and HZE particles to which astronauts are exposed in space, a significant fraction of the cellular changes may be due to bystander responses, in addition to the damages caused in cells directly traversed by ionizing particles or their associated delta rays. We have the following specific aims. 1. Test the hypothesis that bystander effects occur at particle fluences so low that only 1-2 % of cells in a culture dish are traversed by an energetic proton or HZE particle. 2. Extend our low fluence work to studies using repeated exposures of cells to low fluences and mixed (actually sequential) beams of more than one particle type. This will test the hypothesis that cells that have survived an initial traversal and sent a bystander signal, can again send a signal when traversed once again by a particle or can subsequently respond as bystanders to a hit in a neighboring cell. 3. Our studies have indicated that reactive oxygen species (ROS) are at least partially involved in transmittal of the Fe ion-induced bystander signal from hit to unirradiated cells. We will test the hypothesis that non-nuclear damage-initiated, metabolically produced ROS in hit cells are involved in the early steps of bystander signaling. Additionally, we will investigate chemical countermeasures to reduce the bystander responses based on removal of ROS or prevention of their formation. 4. In tissues, cells do not exist in isolation, and there is increasing interest in how signaling between cell types (especially stromal and epithelial) influences cellular responses to stresses. We will test the hypothesis that radiation-induced bystander signals can be sent between differing cell types, but the characteristics of the responses depend on the cell types and the types of radiation.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Although our studies are specifically geared to providing information on the possible biological effects, especially cancer induction, of low fluences of energetic particles on astronauts, the data may have relevance to cancer therapy and the effects of low doses of radiation on radiation workers (e.g., those involved in clean-up of radioactive sites) and the general public. With regards to cancer, there is increasing use of protons (energies of 65-250 MeV) as an effective treatment modality in radiation oncology because they provide a great therapeutic advantage in some types of cancers due to exquisite dose localization. Over 40,000 patients have been treated world-wide with protons, and several new proton centers have opened in the last few years or are under construction in the US. However, despite the clinical success of protons, some concerns exist regarding the possible side-effects, including the possibility of cancer caused by the low doses received by normal tissues. Additionally, there is increasing interest in the use of carbon ions in cancer therapy, with carbon ion treatment centers now in use in Japan and Germany, and the potential being discussed in the US. Our studies could provide important and useful new insight on the effects of these energetic particles on biological systems and the mechanisms by which they act. Such findings would be expected to translate into helping to develop effective ways to reduce possible side-effects, including cancer, from the treatments.		

Task Progress:	Background: Ion-induced media-mediated bystander effects. Previously, we have used a transwell insert co-culture method to demonstrate media-mediated bystander responses in AG01522 normal human fibroblasts after X-irradiation, 1 GeV/n iron ions, and 1 GeV protons. The ion work was performed at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. The work showed that with increasing radiation dose, micronuclei (MN) formation increases in irradiated cells, with the Fe ions being significantly more efficient at producing MN than X-rays or protons. However, in unirradiated bystander cells sharing medium with irradiated cells, the fraction of cells with MN increases about 2-fold at the lowest radiation dose used (10 cGy), then remains constant to the highest dose, irrespective of whether iron ions or protons were used. Similar dose response data for irradiated and bystander cells were obtained using the endpoints of formation of nuclear foci of DNA repair-related proteins, gammaH2AX or 53BP1, as markers of DNA double strand breaks. Although at doses above about 10 cGy the magnitude of the bystander effects is small compared to the damage from direct irradiation, at lower doses the bystander responses make a significant contribution to the overall cell damage measured. Bystander effects induced at very low particle fluences (aim 1). Previously, we reported on our progressing work at NSRL extending studies to very low fluences of 1 GeV/n iron ions and protons. Accomplishing this aim required working closely with Dr. Rusek and his team at NSRL as they developed techniques to monitor the beam and perform dosimetry at very low particle fluences (down to 100 ions/cm^2). Our data show that at fluences where 1% or less of the cells are traversed by an Fe ion, the fraction of cells with MN is the same as in untreated cells (~5%). However, over fluences from ~2e3 to ~4.5e4 (2-50% of cells traversed), MN induction in both irradiated and bystander cells is 1.5-fold higher than in controls. Above 50% cells traversed, MN induction increases with dose in irradiated cells, but remains at the plateau level in bystanders. Similar results occur with 53BP1 foci as the endpoint. Interestingly, with protons, the same level of damaged cells is reached, but the responses shift to higher fluences, i.e., the plateau occurs at 2e5 to 2e7 protons/cm^2 (~2 to 200 protons per cell), above which there is a dose-dependent increase in damage in irradiated cells, but a continued plateau in bystander cells. The data suggest that at low particle fluences, the effects seen in the irradiated population could be due to bystander intercellular signaling, consistent with the observations from Little and colleagues using very low fluences of alpha-particles. Hence, the studies planned for aim 1 with iron ions and protons are complete, and a manuscript has been submitted. We believe that this paper is a unique and important contribution, as it describes data on irradiated and bystander cells at much lower fluences of energetic particles than any other published studies have used to-date. Our finding that about 100-fold lower particle fluence is needed to produce DNA damage in irradiated or bystander cell populations with 1 GeV/n Fe ions than with comparable energy protons suggests that linear energy transfer (LET) (LETs of 1 GeV/n Fe and protons are 151 and 0.24 keV/micrometer, respectively) or track structure is a determinant of the minimum dose needed for an effect. Thus, we have extended the studies planned for this aim to test this theory by using other ions and energies with varying LETs and/or track structures. To date, our data with 300 MeV/n Fe ions (LET 240 keV/micrometer) show elevated fractions of cells with MN in irradiated and bystander populations at 100-800 ions/cm^2, fluences at which MN levels are not elevated with 1 GeV/n Fe. Hence, the data are consistent with the minimum fluence needed for damage induction depending on LET. However, interpretation of the data may not be that simple since just using LET and particle traversals per cell neglects the effects of particle track width from lateral extension of gamma-rays. In recent NSRL campaigns, we have repeated the 300 MeV/n Fe studies and added investigations with 300 and 1000 MeV/n Silicon, 290 MeV/n Carbon and 1000 MeV/n Helium ions (LETs of 77, 44, 13 and 2 keV/micrometer, respectively). We are in the process of analyzing these samples. Bystander responses in “mixed” radiation fields (aim 2). As mentioned in previously, we have been among the pioneering groups at NSRL to use the “mixed”, actually sequential, beams of Fe ions and protons. These studies were designed to assess interactions between the two types of radiation in cells, when the irradiations were separated in time. Such studies are highly relevant to the space environment since it has been estimated that each cell in an astronaut’s body would be traversed by a proton once every three days and by an HZE particle every few months. Our studies have found that sequential exposures to 1 cGy each of 1 GeV protons and 1 GeV/n Fe ions produce the same % cells with MN in the irradiated population as either protons or Fe ions alone. The same magnitude of cells with DNA damage is also observed in non-irradiated bystander cells sharing medium with cells irradiated with either 1 cGy protons or iron ions or protons plus iron ions. However, when the “bystander” cells are exposed to protons up to 3 h before co-culture with Fe ion-irradiated cells, no increase in DNA damage (MN or 53BP1 foci formation) in the “bystander” cells is observed. Hence, this aim has been accomplished and a paper describing the data was published this year. Role of ROS (aim 3). Persistent production of ROS has been shown in mammalian cells after exposure to ionizing radiation, and several studies have reported a role for ROS in bystander signaling. As part of this study, we also investigated whether ROS could mimic the effect of prior proton irradiation in the reduction of the medium-mediated Fe-induced bystander response. Bystander cells were treated with a low level of hydrogen peroxide 30 min, 3 h or 24 h before being put into co-culture with cells irradiated with Fe ions. Treatment with 5 microM hydrogen peroxide alone did not increase the frequency of MN formation, and pretreatment of bystander cells with a low level of hydrogen peroxide did not modulate their ability to show increased MN formation after sharing medium with Fe-irradiated cells. The data imply that ROS may not be involved in the decrease in Fe ion-induced medium-mediated bystander effects caused by a low prior proton dose. These data were also included in the recently published paper. Bystander signaling in mixed cell types and in 3D (aim 4). As mentioned last year, previously obtained preliminary data on signaling between mixed cell types, using AG01522 human fibroblasts and hTERT-immortalized human keratinocytes, had suggested that with 250 kVp X-rays bystander signaling occurs from irradiated fibroblasts to unirradiated keratinocytes and vice versa. However, with 1 GeV/n iron ions, the preliminary data suggested that the bystander response in keratinocytes co-cultured with irradiated fibroblasts is delayed, and no response is seen in the reverse direction. As reported last year, we confirmed those interesting data and extended them, showing that there is a 1.5-2.0-fold increase in MN formation in bystander fibroblasts or keratinocytes after proton irradiation of either cell type, but no significant increase in MN in bystander fibroblasts in co-culture with irradiated keratinocytes. Additional new data now show that a 1.5-2-fold increase in the fraction of cells expressing 53BP1 foci occurs in both cell types when either cell type is irradiated. In bystander cells, the foci occur earlier after proton irradiation (peak about 12 h) than after iron irradiation (peak between 24- 48 h). These data have been presented in poster form, and a manuscript is in preparation. Additional interesting results have shown that there is no bystander signaling with a human chondrosarcoma cell line (HTB194) irradiated with X-rays, protons or iron ions. This is one of the few cell lines in which we have failed to demonstrate a media-mediated bystander effect. However, irradiated chondrosarcoma cells can send signals to bystander AG01522 fibroblasts to cause DNA damage, although the tumor cells do not respond to signals from the fibroblasts. A poster on these data has been presented, and a paper is in preparation. Similar, preliminary data have been obtained with a lung cancer cell line, suggesting that some tumor cells may be resistant to expression of bystander signal-induced damage, although they can send bystander signals. Responses of cells to particle irradiation are reduced in a 3D matrix compared to 2D monolayer. In last year’s report we described our development of a 3D epithelial tissue model in which AG01522 fibroblasts are embedded in a collagen gel with keratinocytes layered on top and grown at an air-liquid interface to enhance keratinocyte differentiation and formation of an epidermal-like layer. We now have this model in routine use and have some interesting findings. The data show that after irradiation with 1 GeV protons, the fraction of cells with DNA damage when irradiated in the 3D constructs is significantly less (4-fold at the shortest time of 1 h) than in cells irradiated in 2D monolayers. With time after irradiation, the fraction of damaged cells decreases such that at 24 h, it is back to background in cells in 3D, but remains elevated out to 72 h in cells in 2D. Published data comparing responses of cells irradiated in 2D and 3D do not all agree; some report that cells in 3D show increased survival and DNA repair, consistent with our findings of decreased level of DNA damage, while others indicate no differences in responses. Bystander responses in cells irradiated in 3D constructs. Using particle microbeam irradiation with MatTek skin constructs, others showed that micronuclei and apoptosis occur in unirradiated (bystander) cells at distances up to 1 mm from the irradiated cells. It is not possible currently at NSRL to irradiate only portions of a 3D construct, so we developed an alternative approach of cutting a construct into two pieces, irradiating half of it, then immediately adding back the unirradiated (bystander) half into the insert containing the irradiated half. Subsequently, both halves are fixed, sectioned and analyzed. As already discussed, in the irradiated constructs, substantial DNA damage is seen 1 h after irradiation, then damage levels decrease with time. After irradiation with protons or Fe ions, 53BP1 foci can be quantified in both keratinocytes and fibroblasts of the bystander 3D construct half. In contrast to the irradiated constructs, the level of damage in the bystander half is low at 1 h, but increases to a peak at 12 h after protons and a maximum at 24-48 h after Fe. The data show that bystander signals can travel considerable distances (mm), consistent with the work of Belykov et al. Bystander signaling in 3D also involves ROS and RNS. Studies by ourselves and others using scavengers have indicted that both ROS and RNS play roles in bystander signaling in cells in monolayer culture. Our recent experiments suggest that the same is also true for cells in 3D. The level of DNA damage in bystander cells in 3D is reduced to control, unirradiated, levels following addition of catalase, SOD or PTIO (scavenging hydrogen peroxide, superoxide and NO, respectively) to the medium being shared by the construct halves. This study has been repeated at recent NSRL runs, with samples yet to be analyzed. We expect that these scavenger data, combined with the data described in the paragraph above on bystander effect in 3D will result in a paper to be prepared as soon as all samples are analyzed.

Bibliography Type:	Description: (Last Updated: 11/13/2018)
Abstracts for Journals and Proceedings	Held KD, Magpayo N, Yang H. "Interactions between targeted and non-targeted effects elicited by combinations of low doses of energetic protons and iron ions." Invited symposium talk presented at the 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, September 2010. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=3cd6ab1e-2dbf-479c-83b0-2b9abbe3eea3&cKey=1002e2aa-4214-47b4-a704-0434ae389563&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Abstracts for Journals and Proceedings	Lumpkins S, Rajadurai C, Yang H, Magpayo N, Held KD. "Comparison of space radiation-induced bystander signaling in 2D and 3D human skin model systems." Poster presented at the 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, September, 2010. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=9e011f35-da29-44e9-81ca-4b52cf617fb4&cKey=d5152a1f-8e35-4ff5-b95f-a6f0f29a0043&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Abstracts for Journals and Proceedings	Wakatsuki M, Magpayo N, Held KD. "The lack of a bystander response induced by photon irradiation in chondrosarcoma cells. " Poster presented at 52nd Annual Meeting of ASTRO (American Society for Radiation Oncology), San Diego, CA, October 2010. Abstracts book for the 52nd Annual Meeting of ASTRO, San Diego, CA, October 2010. , Oct-2010
Abstracts for Journals and Proceedings	Lumpkins SB, Rajadurau, C, Yang H, Magpayo N, Held KD. "Space radiation-induced bystander signaling in a 3D human skin model. " Poster presented at the 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Yang H, Magpayo N, Held KD. "Targeted and non-targeted effects from combinations of low doses of energetic protons and iron ions in human fibroblasts." Int J Radiat Biol. 2011 Mar;87(3):311-9. Epub 2010 Dec 15. PMID: 21158498 , Mar-2011
Articles in Peer-reviewed Journals	Yang H, Magpayo N, Rusek A, Chiang I-H, Sivertz M, Held KD. "Effects of very low fluences of high energy protons or iron ions on irradiated and bystander cells." Radiat Research, submitted, as of June 2011. , Jun-2011